Assessment Tool-1 COT-1 SHORT ANSWER Test Section, Batch : CSE-A1 -2024 Total Marks : 20 Marks. obtained Marks :- 19 Faculty Incharge :- Date : 17/7/26	Name Madhulika RegNo 192472188 course code CSA0702 course Name computer Networks course Faculty Senthamil selvan

course coordinator :-
HOD :-

1) File Transfer using Transport and Data Link Layers.

File size = 150 MB
= 150×10^6 = 150,000,000 bytes

segment size = 1500 bytes.

Number of segments = 150,000,000 ÷ 1500
= 100,000

Number of frames = 100,000

overhead per frame = 40 bytes.

Total overhead = $100,000 \times 40$
= 4,000,000 bytes
= 4MB

Total data transmitted = 154MB

Transmission time = $\frac{154 \times 8}{50}$ = 24.64S

2) Sliding Window Protocol.

Window size = 6

Total frames = 48

Windows required = 48/6 = 8

Retransmission

one frame lost in each window $8 \times 1 = 8$

- Windows Required = 8
- Retransmissions = 8 frames

3) IP Fragmentation.

Packets = 12,000 bytes
MTU = 1500 bytes
IP header = 20 bytes

Payload per fragment = 1500 - 20 = 1480 bytes.
Fragments : [1200/1480] = 9
Header overhead 9x 20 = 180 bytes

- Fragments = 9
- Header overhead = 180 bytes.

4) Sliding window ~~(same~~

Window size = 6

Total frames = 48

windows required = 48/6 = 8

Retransmission.

one frame lost in each window = 8x1=8

windows = 8

Retransmission = 8.

session Layer synchorization

File = 600 MB

check point every 60 MB

Failure after 390 MB

check points : 390/60 = 6.5

completed check points = 6

Last checkpoint = 6x60 = 360 MB

Retransmission:-

390 - 360 = 30 MB

- check points created = 6
- Data retransmitted = 30 MB

6) Presentation Layers compressed and Encryption

original = 900 MB

compression = 40%

Encryption overhead = 5%

compressed size : 900x0.60 = 540 MB

Encrypted size = 540 x 1.05 = 567 MB

Reduction : $\frac{900-567}{900} \times 100 = 37\%$

- compression size = 540 MB
- Encryption size = 567MB
- Total reduction = 37%

7) FTP Data Transfer.

Data = 3GB = 3000MB

Required size = 3MB

Throughput = 120 Mbps

Requests : 3000/3 = 1000

convert data : 3000×8 = 24000 Mbits

Time : 24000/120 = 200 seconds

- Requests = 1000
- Transmission time = 200s.

8) End-to-End Delay.

Layer delays

- Application = 4ms
- Transport = 3ms
- Network = 5ms
- Data link = 2ms
- Physical = 1ms

Processing delay = 4+3+5+2+1 = 15 ms

Propogation : 18ms

Transmission : 12ms

Total : 15+18+12 = 45ms

Total end-to-end delay = 45ms

Frame overhead.

useful data = 80 MB = 80,000,000 bytes
Frames = 1600 bytes
Overhead = 80 bytes
useful Payload: 1600 - 80 = 1520 bytes
Frames: [80,000,000 | 1520] = 52,632
overhead = 52,632 × 80 = 4,210,560 bytes
Efficiency: $\frac{1520}{1600} \times 100 = 95\%$

- Frames = 52,632
- overhead = 4,210,560 bytes
- Efficiency = 95%

10) ## IOT Healthcare Network.

- original = 240MB
- compression = 30%
- Segment = 1400 bytes
- Frame overhead = 60 bytes

compressed size 240 × 0.70 = 168 MB
compressed bytes 168,000,000 bytes
Segments 168,000,000/1400 = 120,000
Protocol overhead : 120,000 × 60 = 7,200,000 bytes = 7.2MB
Final transmitted : 168 + 7.2 = 175.2MB

Q.L.T

understanding concepts	Explanation	Application	organization	Accuracy	clarity
25%	25%	20%	15%	10%	5%
5	5	4	3	1	1